

SWARNALI MOLLICK

JUNIOR DATA SCIENTIST | MACHINE LEARNING ENGINEER | APPLIED AI ENGINEER

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PROFESSIONAL SUMMARY

AI and Data Engineering MSc candidate with hands-on experience building end-to-end data science, machine-learning, NLP and generative-AI systems. Strong in Python, SQL, Pandas, feature engineering, statistical analysis, model comparison, business interpretation and deployment. Delivered classification, forecasting, transformer and RAG applications using Scikit-learn, PyTorch, Hugging Face, FastAPI, Docker and Streamlit. Research background includes peer-reviewed work in healthcare AI, environmental prediction and data mining. Seeking junior Data Scientist, Machine Learning Engineer or Applied AI Engineer opportunities in Europe.

TECHNICAL SKILLS

Programming & Data: Python, SQL, MySQL, R, Pandas, NumPy, PySpark, Apache Spark, data cleaning, EDA, feature engineering, statistical analysis

Machine Learning: Scikit-learn, XGBoost, Logistic Regression, Random Forest, classification, regression, forecasting, cross-validation, imbalanced-data analysis, model evaluation

Deep Learning & NLP: PyTorch, TensorFlow/Keras, Transformers, DistilBERT, Hugging Face, LangChain, RAG, embeddings, ChromaDB, prompt engineering, semantic search

Deployment & Tools: FastAPI, Docker, Streamlit, Flask, Git/GitHub, Jupyter Notebook, VS Code, APIs, technical documentation

Visualization: Matplotlib, Seaborn, Plotly, data storytelling, business insights

MACHINE LEARNING & AI PROJECTS

Customer Churn Prediction - Machine Learning API | [GitHub](#)

- Built an end-to-end churn pipeline for 7,043 telecom customers, covering EDA, preprocessing, feature engineering and comparison of Logistic Regression, Random Forest and XGBoost.
- Achieved ROC-AUC 0.8454 and accuracy 0.8041 with Logistic Regression; identified contract type, payment method, technical support and online security as key churn drivers.
- Translated findings into retention recommendations and deployed the preprocessing and prediction workflow through a FastAPI REST endpoint and Docker container.

Retail Sales Prediction and Business Analytics | [GitHub](#)

- Analyzed 1,017,209 Rossmann retail records to quantify the effects of promotions, holidays, seasonality and store characteristics; promotions increased average sales by approximately 39%.
- Compared Linear Regression, Decision Tree, Gradient Boosting, XGBoost and Random Forest; the best model achieved R^2 0.868, MAE 740.34 and RMSE 1,128.59.
- Converted feature-importance and EDA findings into recommendations for inventory, staffing and promotional planning.

Financial Sentiment Classification & Streamlit App | [GitHub](#)

- Designed a three-class financial NLP experiment comparing fine-tuned DistilBERT with few-shot Llama 3.1 prompting for Bearish, Bullish and Neutral text classification.
- DistilBERT achieved approximately 85.8% accuracy and 82.0% Macro-F1, outperforming the prompted LLM while providing substantially lower inference latency.
- Selected the smaller specialized model based on accuracy, latency and deployment cost, then built a Streamlit prediction application for practical use.

AI/NLP Learning Assistant - Retrieval-Augmented Generation | [GitHub](#)

Live demo: ai-nlp-learning-assistant.streamlit.app

- Built a RAG assistant using LangChain, ChromaDB, Hugging Face embeddings and Llama 3.1 through the Groq inference API to retrieve learning materials and generate context-grounded responses.
- Integrated intent classification, document ingestion, semantic search and conversation history; deployed the application on Streamlit Cloud.
- Resolved vector-store configuration, caching, loader, dependency and retrieval-chain integration issues during cloud deployment.

PROFESSIONAL EXPERIENCE

Web Developer / Web Programmer - Appstick, Bangladesh

Sep 2022 - Sep 2023

- Developed and maintained responsive, database-backed web applications using HTML, CSS, Bootstrap, Laravel, PHP and MySQL.
- Implemented MySQL-driven features, maintained structured data flow across frontend, backend and database layers, and documented development workflows.

Web Designer Intern - The Virtual BD, Bangladesh

Apr 2021 - Feb 2022

- Designed responsive WordPress and Elementor pages, implemented HTML-based customizations and received the Best Web Designer Award for internship performance.

EDUCATION

MSc in Computer Science and Information Technology (AI and Data Engineering)

Sep 2024 - Dec 2026

Ca' Foscari University of Venice, Italy | 90 ECTS completed; Current weighted average: 27.13/30

- Relevant coursework: Foundations of Machine Learning, Foundations of Artificial Intelligence, Advanced Data Management, Calculus and Optimization, Image and Video Understanding, Deep Learning for NLP and Time Series for Computer Science.

BSc in Computer Science and Engineering

2018 - 2022

Northern University of Business and Technology Khulna, Bangladesh | CGPA 3.959/4.00 | 152 credits

- Undergraduate thesis: Adapted Deep Ensemble Learning-Based Voting Classifier for Osteosarcoma Cancer Classification.

RESEARCH & SELECTED PUBLICATIONS

Co-author of 10+ journal and conference publications in healthcare AI, machine learning, data mining, association-rule mining and environmental prediction. Full record: [Google Scholar](#)

- Adapted deep ensemble learning-based voting classifier for osteosarcoma cancer classification - Diagnostics, 2023.
- A comparative study, prediction and development of chronic kidney disease using machine learning on patient clinical records - Human-Centric Intelligent Systems, 2023.
- An unsupervised cluster-based feature grouping model for early diabetes detection - Healthcare Analytics, 2022.
- An Apriori algorithm-based association-rule analysis to detect human suicidal behaviour - Procedia Computer Science, 2022.
- Efficient prediction of water quality index using machine-learning algorithms - Human-Centric Intelligent Systems, 2021; Best Paper Award.

HONOURS & AWARDS

- Italian Government MAECI Scholarship, 2024-2026.
- Best Paper Award, Human-Centric Intelligent Systems, Springer, 2022.
- Best Web Designer Award, The Virtual BD, 2021.
- NUBTK Departmental Scholarship for Excellent Result, 2018 and 2019.

COURSES & CERTIFICATIONS

IBM/ Coursera: What is Data Science? | **IBM/ Coursera:** Machine Learning with Python | **aiQuest:** Data Science and Machine Learning with Python | **Ostad:** Data Science Certificate Program | **MathWorks:** MATLAB Onramp | **HackerRank:** Python Basic

ACADEMIC SERVICE, LEADERSHIP & LANGUAGES

Academic service: Conference Reviewer, International Conference on Computer Science and Application Engineering (CSAE), 2023; IEEE Student Member, 2022-2023.

Leadership: Class Representative, NUBTK; Programming Club member who organized a programming competition and delivered AI-focused talks to junior students.

Languages: Bengali (native), English (IELTS 7.0), Italian (A1, actively learning), Hindi (conversational).